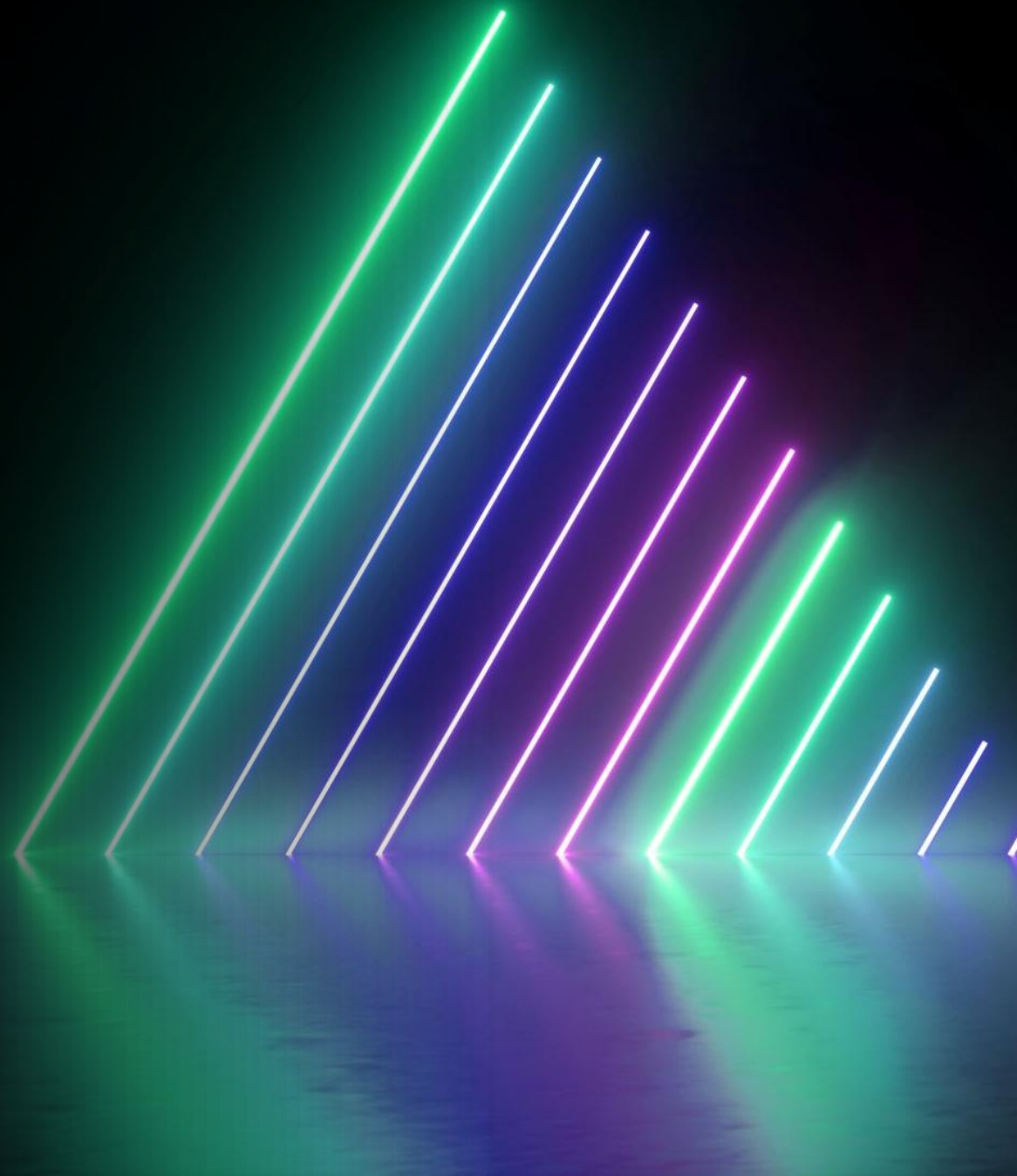




Getting Started with R

Olayinka I. Arimoro





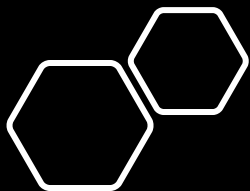
Meet Olayinka Arimoro

- Seven (7) years of experience using R and RStudio.
- 4+ years of experience building machine learning models working with health data.
- Top-rated Coursera guided project instructor with 40 courses and 100K+ learners.
- Background in Mathematics and Education; Statistics; Biostatistics
- 2nd year PhD student, Biostatistics, University of Calgary
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R Overview & Preliminaries



What we will cover today

What is R?

Why learn R?

Installing R and RStudio

Basic data types in R

What is R?

- R is not just the letter of the alphabet.
- R is a language and environment for statistical analysis, computing and graphics representation, and reporting.
- R is based heavily on the S language, first developed in the 1960s and 1970s by researchers at Bell Laboratories in New Jersey.
- R was created by Ross Ihaka and Robert Gentleman at the University of Auckland, New Zealand, to embrace open-source software.
- R is an open-source software for statistical & data science programming and one of the most popular, powerful, and reliable tools.

Why Learn R?

- R is a popular language in the world of Data Science.
- It is heavily used in analyzing data that is both structured and unstructured.
- Free and Open-source tool
- Large community of users
- Latest cutting-edge technology
- Independent Platform
- Gateway to lucrative career

Why Learn R?

- The right tool for (many of our) jobs
- Used in almost every industry
- **Powerful:** Gives access to CRAN repository containing over 19,000 packages with pre-defined functions for almost every purpose
- Reproducible, sharable codes & tools for collaboration
 - Keep in mind: Every analysis you do on a dataset may have to be redone more than once (probably 3 – 15 times) before publication. So, make sure your code is reproducible!

Common uses of R

- **Exploratory analyses:** statistical summaries, plots or statistical analyses we learned – t-test, ANOVA, linear regression, and so on.
- **Advanced analyses:** Bayesian inference, mixed effects models, random forests, tree-based models, deep learning models, and so on.
- Generate publication-quality figures or interactive figures
- Build your own R packages and share via repositories, e.g., GitHub
- Build interactive platforms using R Shiny, e.g., COVID-19 tracker

Learning R



R is a programming language, meaning the learning curve can be steeper, so be patient.



Increased productivity when fluent

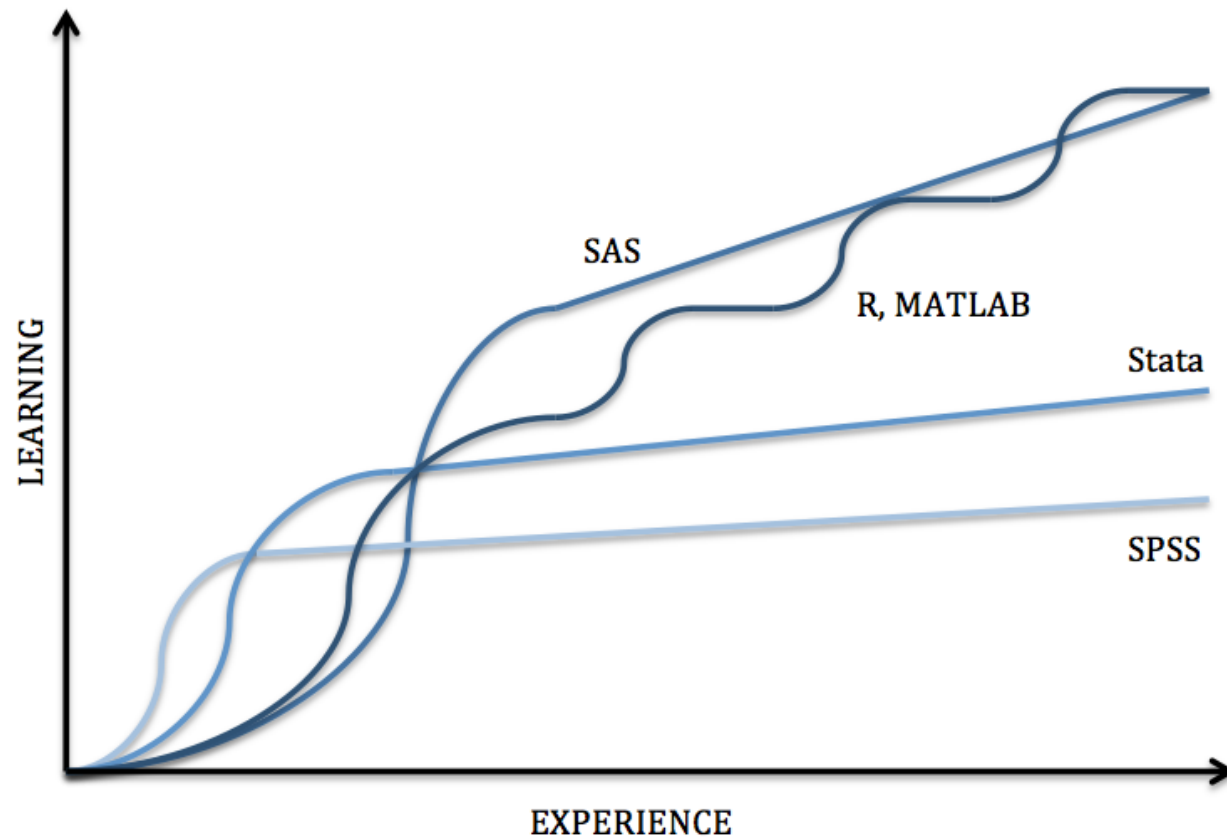


<https://sites.google.com/a/nyu.edu/statistical-software-guide/summary>

Note:

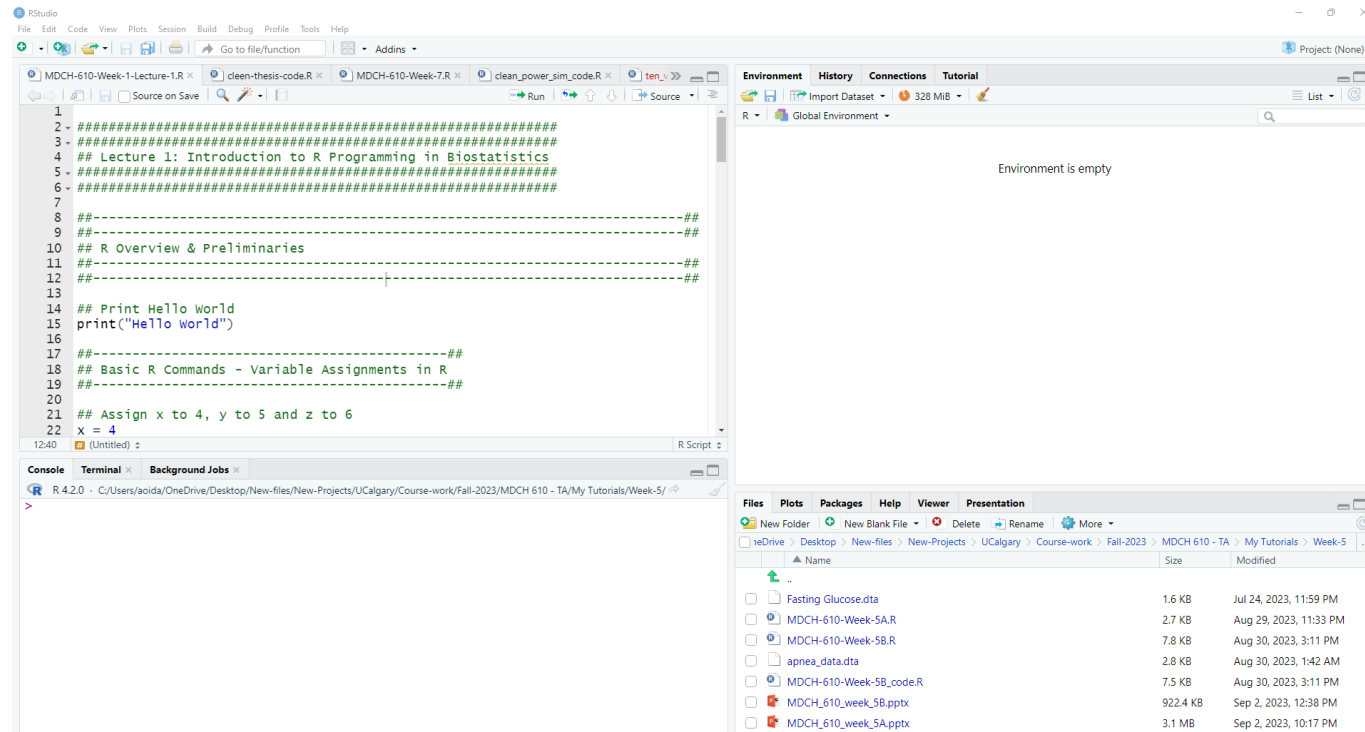
- You learn to program by making mistakes
- Expect to make errors, learn from them, avoid them but do not let errors frustrate you.

Learning R



What is RStudio?

- Portal through which to use R – Integrated Development Environment (IDE)
- You can write code (in a script), execute code line by line, manage data, get help, view plots, and so on.





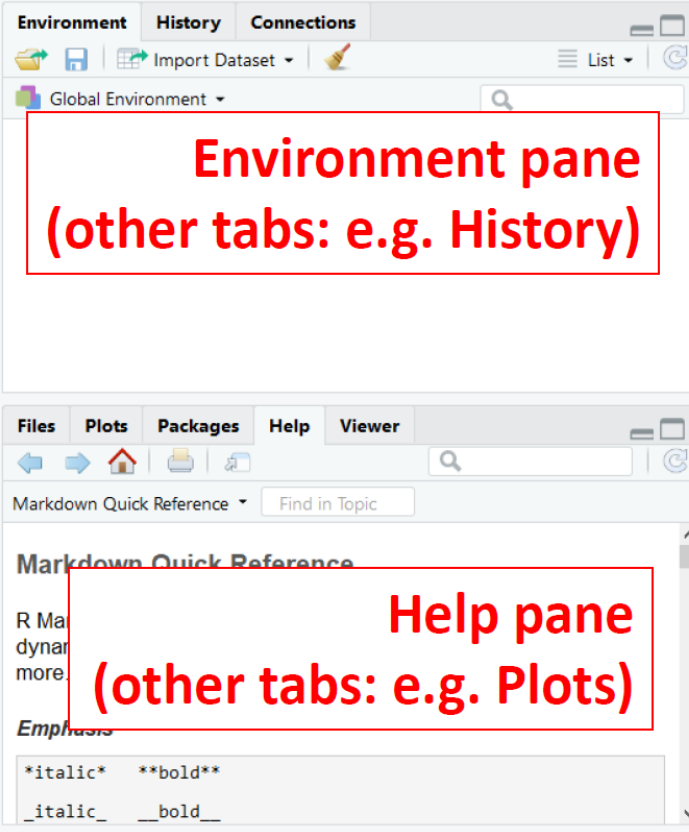
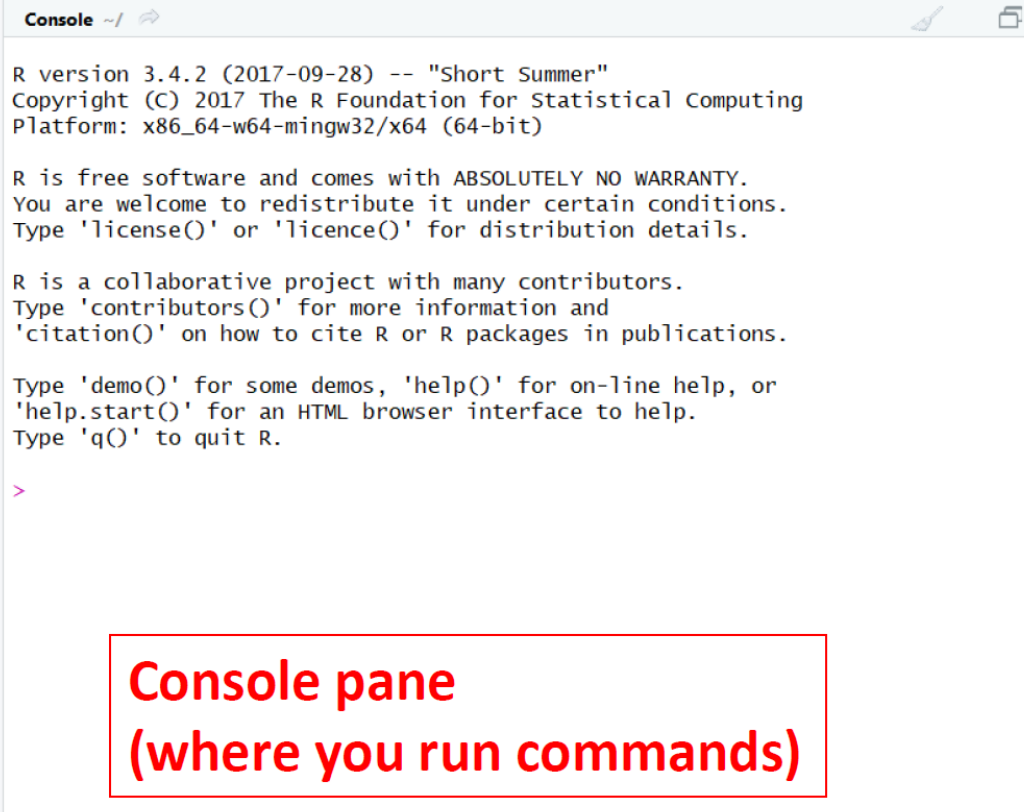
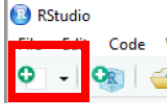
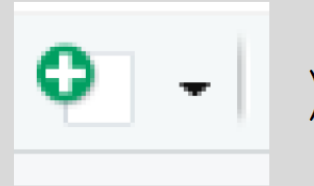
Let's download and install R and RStudio

- Download R for Windows and macOS: <https://cran.r-project.org/>
- Download RStudio Desktop: <https://posit.co/download/rstudio-desktop/>
- Installing R and RStudio: <https://rstudio-education.github.io/hopr/starting.html>
- Install R and RStudio on Mac: <https://teacherscollege.screenstepslive.com/a/1135059-install-r-and-r-studio-for-mac>

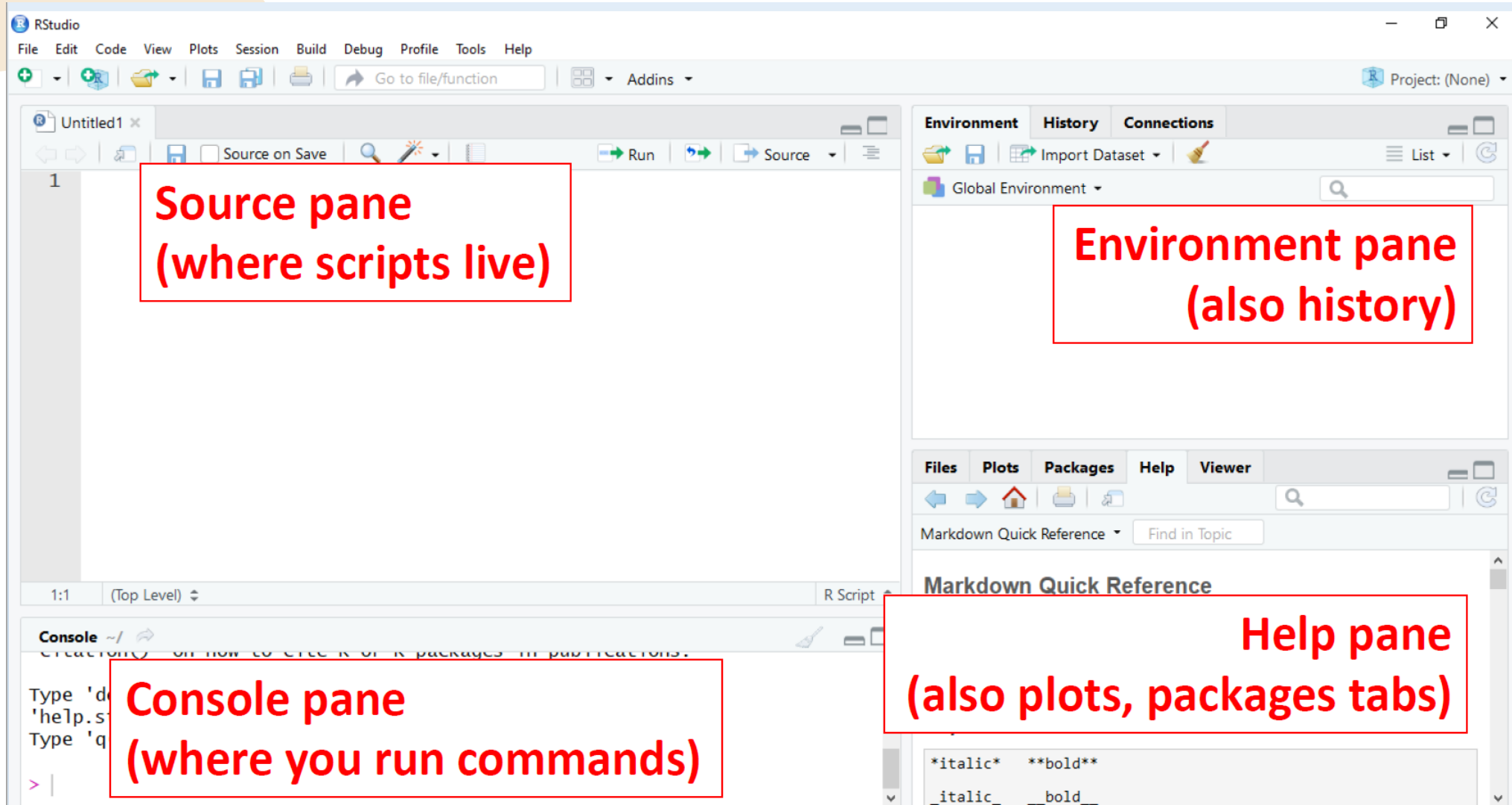
RStudio: IDE for R

Instruction:

- Click on the Green Plus (i.e
- Then, click on R script



RStudio: IDE for R



RStudio – The Source Pane

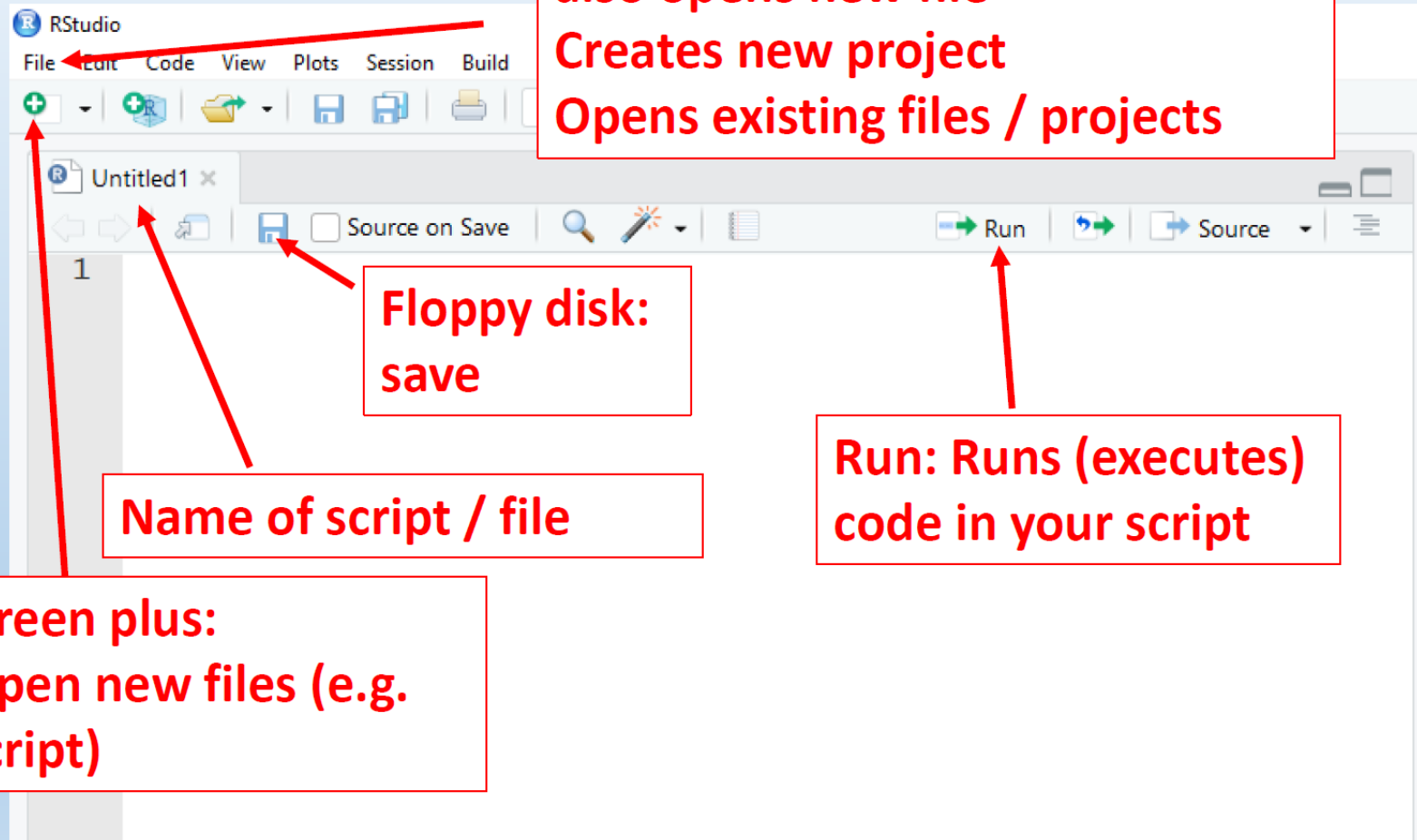
- Scripts (your code)

File:

also opens new file

Creates new project

Opens existing files / projects



**Floppy disk:
save**

Name of script / file

**Run: Runs (executes)
code in your script**

**Green plus:
Open new files (e.g.
script)**

RStudio – The Plots/Packages/Help Pane

- **Option 1:** Type in Help pane search bar

Type log into search bar

- **Option 2:** Type in Console, examine in Help pane

Type in console

```
> ?anova
```

Examine output in Help pane

- **Option 3:** Type in Console, examine in Help pane

Type in console

```
> ??"analysis of variance"
```

Examine in Help pane

log (base)

R Documentation

Logarithms and Exponentials

Description

What the function does in general terms

`log` computes logarithms, by default natural logarithms, `log10` computes common (i.e., base 10) logarithms, and `log2` computes binary (i.e., base 2) logarithms. The general form `log(x, base)` computes logarithms with base `base`.

`log1p(x)` computes $\log(1+x)$ accurately also for $|x| \ll 1$ (and less accurately when x is approximately -1).

`exp` computes the exponential function.

`expm1(x)` computes $\exp(x) - 1$ accurately also for $|x| \ll 1$.

Usage

How to use the function

```
log(x, base = exp(1))
logb(x, base = exp(1))
log10(x)
log2(x)
```

```
log1p(x)
```

```
exp(x)
expm1(x)
```

Arguments

What does the function need

`x` a numeric or complex vector.

`base` a positive or complex number: the base with respect to which logarithms are computed. Defaults to $e = \exp(1)$.

Details

All except `logb` are generic functions: methods can be defined for them individually or via the [Math](#) group generic.

`log10` and `log2` are only convenience wrappers, but logs to bases 10 and 2 (whether computed via `log` or the wrappers) will be computed more efficiently and accurately where supported by the OS. Methods can be set for them individually (and otherwise methods for `log` will be used).

`logb` is a wrapper for `log` for compatibility with S. If (S3 or S4) methods are set for `log` they will be dispatched. Do not set S4 methods on `logb` itself.

All except `log` are [primitive](#) functions.

Value

What does the function return

A vector of the same length as `x` containing the transformed values. `log(0)` gives $-\text{Inf}$, and `log(x)` for negative values of `x` is NaN . `exp(-Inf)` is 0.

For complex inputs to the log functions, the value is a complex number with imaginary part in the range $[-\pi, \pi]$: which end of the range is used might be platform-specific.

S4 methods

`exp`, `expm1`, `log`, `log10`, `log2` and `log1p` are S4 generic and are members of the [Math](#) group generic.

Note that this means that the S4 generic for `log` has a signature with only one argument, `x`, but that `base` can be passed to methods (but will not be used for method selection). On the other hand, if you only set a method for the `Math` group generic then `base` argument of `log` will be ignored for your class.

Source

`log1p` and `expm1` may be taken from the operating system, but if not available there are based on the Fortran subroutine `dlnrel` by W. Fullerton of Los Alamos Scientific Laboratory (see <http://www.netlib.org/slatec/fnlib/dlnrel.f> and (for small `x`) a single Newton step for the solution of $\log1p(y) = x$ respectively.

References

Becker, R. A., Chambers, J. M. and Wilks, A. R. (1988) *The New S Language*. Wadsworth & Brooks/Cole. (for `log`, `log10` and `exp`.)

Chambers, J. M. (1998) *Programming with Data. A Guide to the S Language*. Springer. (for `logb`.)

See Also

Discover other related functions

[Trig](#), [sqrt](#), [Arithmetic](#).

Examples

Sample code showing how it works

```
log(exp(3))
log10(1e7) # = 7

x <- 10^-(1+2*1:9)
cbind(x, log(1+x), log1p(x), exp(x)-1, expm1(x))
```

Other Ways to Get Help in R

- Search documentation: `help.search("keyword")`
- Check the package vignettes: `browseVignettes("grmtree")`
- Online resources:
 - A simple Google search
 - CRAN: <https://cran.r-project.org/>
 - F-bloggers: <https://www.r-bloggers.com/>
 - Stack Overflow: <https://stackoverflow.com/>

Shortcut to Access the Interface

- Ctrl + 1 : Source editor
- Ctrl + 2 : Console
- Ctrl + 3 : Help
- Ctrl + 4 : History
- Ctrl + 5 : Files
- Ctrl + 6 : Plots
- Ctrl + 7 : Packages
- Ctrl + 8 : Environment
- Ctrl + 9 : Viewer
- To view any interface in full-screen, add **shift**. E.G., Ctrl + Shift + 1: Source editor in full screen

Basic Data Types in R



Basic Data Types in R

R works with numerous data types. The most common types are:

- Decimals values like 3.5, called **'numeric'**
- Natural numbers like 3 are called **'integers'**. *Integers are also numeric*
- Boolean variables (TRUE or FALSE) are classified as **'logical'**
- Text (or string) values are classified as **'character'**
- Categorical variables are called **'factors'**. They have a finite and defined set of values they can take (e.g. eye_color can have a value contained in {'blue', 'green', 'brown', 'black'})

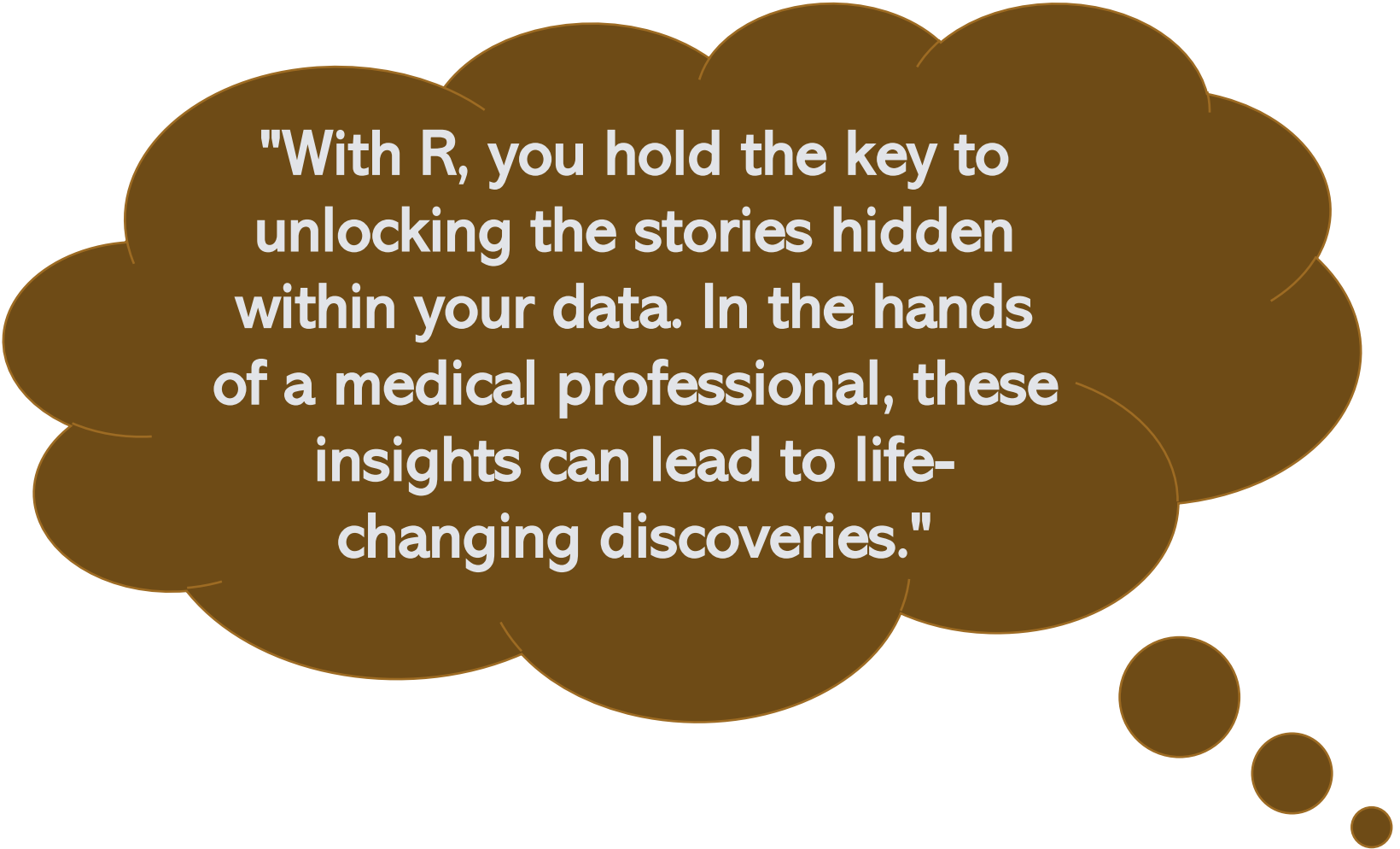
Working Directory

- **Definition:** The working directory is the folder where R looks for files to read and where it saves files.
- **Setting the Working Directory:**
 - Using RStudio: **Session > Set Working Directory > Choose Directory.**
 - Using code: **setwd("path/to/your/folder")**
- Checking the current working directory: **getwd()**
- Shortcut: **Ctrl + Shift + H (recommended)**

Final Thoughts

- Explore the RStudio software; don't be afraid to make mistakes
- Always remember to set your working directory
- Use comments when coding to improve reproducibility
- Remember that RStudio is case sensitive
- Make use of shortcuts to save time

Final Thoughts



"With R, you hold the key to unlocking the stories hidden within your data. In the hands of a medical professional, these insights can lead to life-changing discoveries."

Resources

- <https://www.r-project.org/about.html>
- <https://data-flair.training/blogs/why-learn-r/>
- <https://blog.imarticus.org/how-important-is-the-r-programming-language-nowadays-machine-learning-blog/>
- <https://prutor.ai/why-r-is-worth-learning/>
- **Important:** <https://cran.r-project.org/doc/manuals/r-release/R-lang.pdf>

**to be
continued...**

thanks for listening.

Thank You

